

OOPS ASSIGNMENT

Question1.

Code :

```
#include<iostream>

using namespace std;

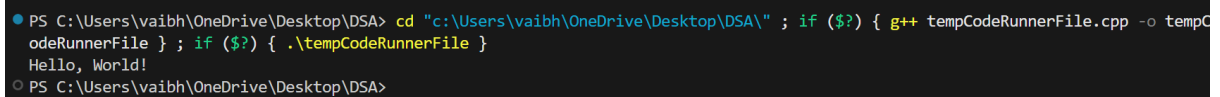
int main() {

    cout << "Hello, World!" << endl;

    return 0;

}
```

Output:



```
PS C:\Users\vaibh\OneDrive\Desktop\DSA> cd "c:\Users\vaibh\OneDrive\Desktop\DSA\" ; if ($?) { g++ tempCodeRunnerFile.cpp -o tempC
odeRunnerFile } ; if ($?) { .\tempCodeRunnerFile }
Hello, World!
PS C:\Users\vaibh\OneDrive\Desktop\DSA>
```

Question2.

Code:

```
#include<iostream>

using namespace std;

int main() {

    int x;

    float y;

    char z;

    cin >> x >> y >> z;

    cout << "Integer: " << x << endl;

    cout << "Float: " << y << endl;

    cout << "Character: " << z << endl;

    return 0;

}
```

Output:

```
PS C:\Users\vaibh\OneDrive\Desktop\DSA> cd "c:\Users\vaibh\OneDrive\Desktop\DSA\" ; if ($?) { g++ assignment.cpp -o assignment } ; if
?) { .\assignment }
1
1.1
a
Integer: 1
Float: 1.1
Character: a
PS C:\Users\vaibh\OneDrive\Desktop\DSA> |
```

Question3.

Code:

```
#include<iostream>

using namespace std;

int main() {

    int x, y;

    cout << "Enter numbers : ";

    cin >> x >> y;

    int a = x + y;

    int b = x - y;

    int c = x * y;

    int d = x / y;

    cout << "Addition: " << a << endl;

    cout << "Subtraction: " << b << endl;

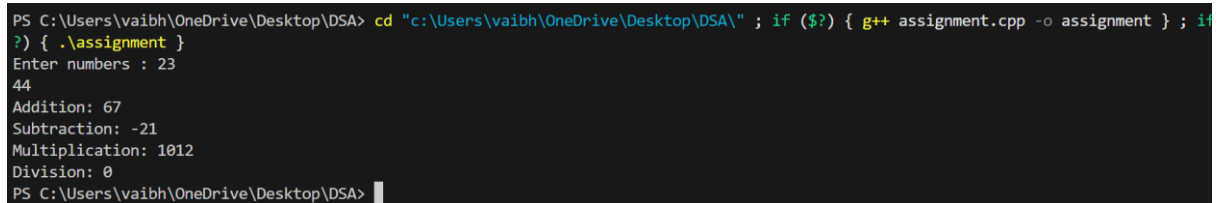
    cout << "Multiplication: " << c << endl;

    cout << "Division: " << d << endl;

    return 0;

}
```

Output:



```
PS C:\Users\vaibh\OneDrive\Desktop\DSA> cd "c:\Users\vaibh\OneDrive\Desktop\DSA\" ; if ($?) { g++ assignment.cpp -o assignment } ; if
?) { .\assignment }
Enter numbers : 23
44
Addition: 67
Subtraction: -21
Multiplication: 1012
Division: 0
PS C:\Users\vaibh\OneDrive\Desktop\DSA>
```

Question4.

Code:

```
#include<iostream>

using namespace std;

int main() {

    int C;

    cout<< "Enter the temperature in Celsius: ";

    cin>>C;

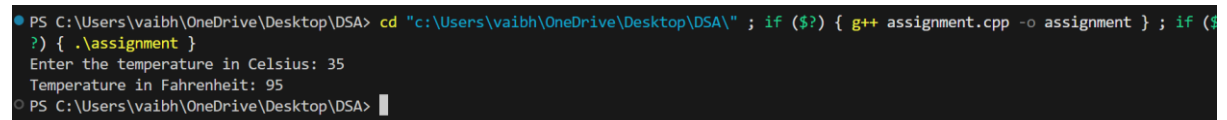
    float F = (C * 9/5) + 32;

    cout << "Temperature in Fahrenheit: " << F << endl;

    return 0;

}
```

Output:



```
PS C:\Users\vaibh\OneDrive\Desktop\DSA> cd "c:\Users\vaibh\OneDrive\Desktop\DSA\" ; if ($?) { g++ assignment.cpp -o assignment } ; if ($?) { .\assignment }
Enter the temperature in Celsius: 35
Temperature in Fahrenheit: 95
PS C:\Users\vaibh\OneDrive\Desktop\DSA>
```

Question5.

Code:

```
#include<iostream>

#include<iomanip>

using namespace std;

int main() {

    cout << "Hello, World!\n" ;

    cout << "Hello, World!\t" ;

    cout << "Hello, World!\b" ;

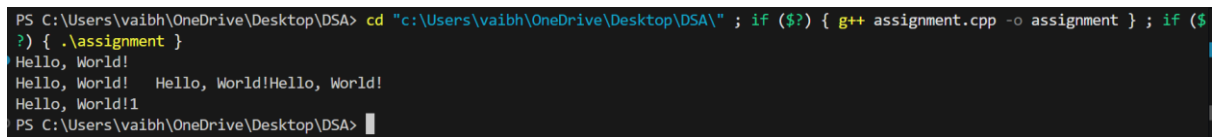
    cout << "Hello, World!" << endl ;

    cout << "Hello, World!" << setw ;

    return 0;

}
```

Output:



```
PS C:\Users\vaibh\OneDrive\Desktop\DSA> cd "c:\Users\vaibh\OneDrive\Desktop\DSA" ; if ($?) { g++ assignment.cpp -o assignment } ; if ($?) { .\assignment }
Hello, World!
Hello, World! Hello, World!Hello, World!
Hello, World!1
PS C:\Users\vaibh\OneDrive\Desktop\DSA>
```

Question 6:

Code:

```
#include <iostream>

using namespace std;

int main() {

    int a = 10;

    a += 5;

    cout << "After += 5, a = " << a << endl;

    a -= 3;

    cout << "After -= 3, a = " << a << endl;

    return 0;

}
```

Output:

```
PS C:\Users\vaibh\OneDrive\Desktop\DSA> cd "c:\Users\vaibh\OneDrive\Desktop\DSA" ; if ($?) { g++ assignment.cpp -o assignment } ; if ($?) { .\assignment }
After += 5, a = 15
After -= 3, a = 12
PS C:\Users\vaibh\OneDrive\Desktop\DSA>
```

Question7.

Code:

```
#include <iostream>

using namespace std;

int main() {

    int a, b;

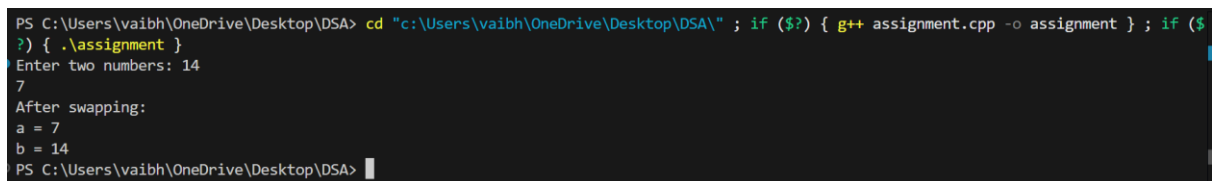
    cout << "Enter two numbers: ";
    cin >> a >> b;

    a = a ^ b;
    b = a ^ b;
    a = a ^ b;

    cout << "After swapping:" << endl;
    cout << "a = " << a << endl;
    cout << "b = " << b << endl;

    return 0;
}
```

Output:



```
PS C:\Users\vaibh\OneDrive\Desktop\DSA> cd "c:\Users\vaibh\OneDrive\Desktop\DSA\" ; if ($?) { g++ assignment.cpp -o assignment } ; if ($?) { .\assignment }
Enter two numbers: 14
7
After swapping:
a = 7
b = 14
PS C:\Users\vaibh\OneDrive\Desktop\DSA>
```

Question8.

Code:

```
#include <iostream>

using namespace std;

int main() {
    int days;
    float fine = 0;

    cout << "Enter number of days late: ";
    cin >> days;

    if (days <= 5) {
        fine = days * 0.50;
        cout << "Fine = Rs. " << fine << endl;
    }
    else if (days <= 10) {
        fine = 5 * 0.50 + (days - 5) * 1;
        cout << "Fine = Rs. " << fine << endl;
    }
    else if (days <= 30) {
        fine = 5 * 0.50 + 5 * 1 + (days - 10) * 5;
        cout << "Fine = Rs. " << fine << endl;
    }
    else {
        cout << "Membership cancelled." << endl;
    }

    return 0;
}
```

Output:


```
PS C:\Users\vaibh\OneDrive\Desktop\DSA> cd "c:\Users\vaibh\OneDrive\Desktop\DSA\" ; if ($?) { g++ assignment.cpp -o assignment } ; if ($?) { .\assignment }  
Enter number of days late: 8  
Fine = Rs. 5.5  
PS C:\Users\vaibh\OneDrive\Desktop\DSA>
```

Question9.

Code:

```
#include <iostream>
```

```
using namespace std;
```

```
int main() {
```

```
    int a, b, choice;
```

```
    cout << "Enter two numbers: ";
```

```
    cin >> a >> b;
```

```
    cout << "1. Add\n2. Subtract\n3. Multiply\n4. Divide\n";
```

```
    cout << "Enter choice: ";
```

```
    cin >> choice;
```

```
    switch (choice) {
```

```
        case 1:
```

```
            cout << "Addition = " << a + b;
```

```
            break;
```

```
        case 2:
```

```
            cout << "Subtraction = " << a - b;
```

```
            break;
```

```
        case 3:
```

```
            cout << "Multiplication = " << a * b;
```

```
            break;
```

```
        case 4:
```

```
            cout << "Division = " << a / b;
```

```
            break;
```

default:

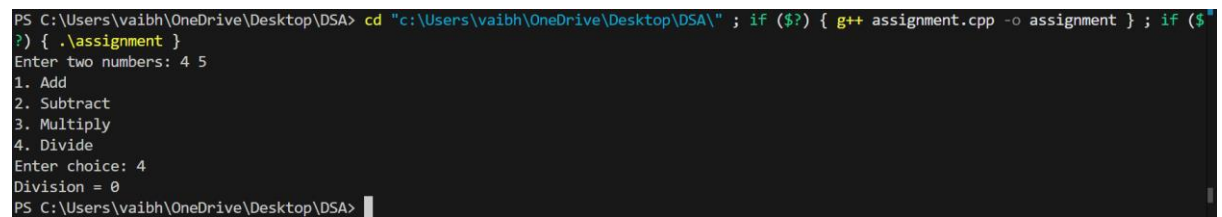
```
cout << "Invalid choice";
```

```
}
```

```
return 0;
```

```
}
```

Output:



```
PS C:\Users\vaibh\OneDrive\Desktop\DSA> cd "C:\Users\vaibh\OneDrive\Desktop\DSA\" ; if ($?) { g++ assignment.cpp -o assignment } ; if ($?) { .\assignment }
Enter two numbers: 4 5
1. Add
2. Subtract
3. Multiply
4. Divide
Enter choice: 4
Division = 0
PS C:\Users\vaibh\OneDrive\Desktop\DSA>
```

Question10.

Code:

```
#include <iostream>
```

```
using namespace std;
```

```
int main() {
```

```
    int n;
```

```
    cout << "Enter a number: ";
```

```
    cin >> n;
```

```
    if (n % 2 == 0)
```

```
        cout << "Using if-else: Even" << endl;
```

```
    else
```

```
        cout << "Using if-else: Odd" << endl;
```

```
    (n % 2 == 0) ? cout << "Using ?: Even" : cout << "Using ?: Odd";
```

```
    return 0;
```

```
}
```

Output:

```
PS C:\Users\vaibh\OneDrive\Desktop\DSA> cd "c:\Users\vaibh\OneDrive\Desktop\DSA\" ; if ($?) { g++ assignment.cpp -o assignment } ; if ($?) { .\assignment }
Enter a number: 5
Using if-else: Odd
Using ?: Odd
PS C:\Users\vaibh\OneDrive\Desktop\DSA>
```